Regression Analysis

Regression Analysis is a statistical and machine Learning method used to :

→ understand relationships between variables
→ Measure how strongly they are related
→ Predict future numerical values
→ Explain how changes in one variable effect another

It does this by building a mathematical model (line / curve) between :

→ Dependent Variable (Y) → the value you want to predict
→ Independent Variable ($X_1, X_2, \ldots\ldots$) → factors affecting y.

Flow of Regression Analysis

i) Define the research questions
ii) Collect and prepare data
iii) Visualize the data
iv) Check assumptions :- L.R has some assumptions, including linearity, independence of errors, homoscedasticity (constant Variance of errors) and normality of Errors
→ You can check /use diagnostic plots and statistical tests to check whether these assumptions hold for your data
v) fit the linear Regression model
vi) Interpret the model } Regression analysis will be done here
→ Analyze the estimated regression coefficients, their standard errors, t-value and p-values to determine the statistical significance of the relationship b/w dependent & independent variables
→ R2 score, adjusted R2-score value can provide insights into the goodness-of-fit of the model
vii) validate the model (split into train, test fit, predict)
viii) Report results

Why ML problems are a Statistical Inference Problems :-

Every machine Learning algorithm learns patterns from data and then applies them to new unseen data.

Machine learning doesn't predict randomly
It transfers knowledge / statistics from sample data.

Inference = Learn pattern from Sample and generalize

Ex:- Train model on 1,000 houses in a particular city
→ It infers that larger houses usually cost more
→ Then it applies this learning to new houses.

ML uses Statistical Inference because:-
- we use sample data → Real world is too large to measure fully.
- We Estimate relationships → Exact values are not known
- we minimize error → based on probability & statistics
- we generalize to future data → Prediction = inference

Ex:-

Cgpa	iq	lpa
-	-	-
-	-	-
-	-	-

Sample data

$\boxed{lpa = f(cgpa, iq)}$ ← model predicts based on this

but there will be some noise / factor which we cannot Explain in terms of maths Equation

$$\boxed{lpa = f(cgpa, iq) + \varepsilon}$$

(irreducible Error)

$$\boxed{lpa = \underbrace{\beta_0 + \beta_1 cgpa + \beta_2 iq}_{\text{population}}}$$ } ml, LR model assumes this Eq

Sample { $\boxed{b_0, b_1, b_2} \neq \boxed{\beta_0, \beta_1, \beta_2}$

{ $\underbrace{f() - f'()}_{\text{reducable}}$

$$lpa = f'(cgpa, iq) + \boxed{\text{reducable}} + \underset{(\text{irreducable error})}{\varepsilon}$$

$f() \simeq f'()$

$f'()$ estimation values of x & y for given data

$f()$ ⇒ True values of X, Y for Population { not possible

*: for Interview

Inference vs Prediction [Why regression analysis is required

Inference means understanding relationships in data you are not just predicting, you want to know why & How one variable affects another

→ Inference focuses on understanding cause / effect

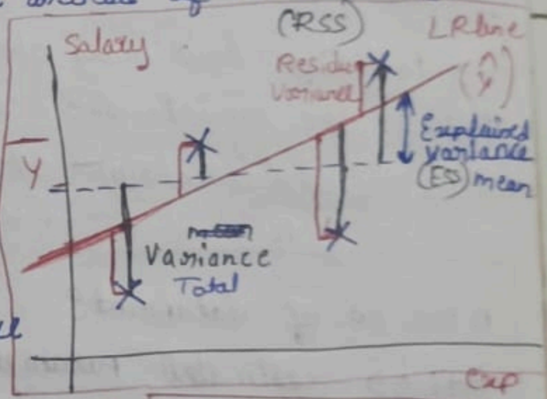
Feature	Prediction	Inference
Goal	Guess the future Values	understand relationships

• what it answers	→ wt will happen	→ why does it happen
• ML models best for	→ Random forest, Neural Networks, XGBoost	→ Regression, statistical model
• Focus	→ Accuracy	→ Explanation
Ex:-	→ Predict house price	→ know how location, size affect price

→ Prediction is about accuracy
→ Inference is about understanding

regression analysis is required because:-

	Purpose
• To infer relationships	→ Understand how variables affect outcomes
• To make prediction	→ predict future values based on past data.
• To measure impact	→ Quantify how much change happens
• To estimate uncertainity	→ Provide statistically confident result
• To make decisions	→ Use data instead of assumptions



TSS, RSS and ESS :-

TSS → Total sum of squares
RSS → Residual sum of squares
ESS → Explained sum of squares

$$\boxed{TSS = \Sigma(y_i - \bar{y})^2}$$ overall variance in data

$$\boxed{RSS = \Sigma(y_i - \hat{y})^2}$$ unExplained Variance in data

$$\boxed{ESS = TSS - RSS}$$ Explained variance in the data

$$\boxed{TSS = \underbrace{ESS}_{\text{Explained}} + \underbrace{RSS}_{\text{unExplained}}} \rightarrow (\varepsilon)$$

unExplained due to → reducible Error, → irreducible Error (ε)

Degree of Freedom :-

→ Degree of freedom (DOF) tells us how many independent values are available in your data to estimate a statistical parameter (like mean, variance, regression coefficients etc)
→ It represents how many values are free to vary.

Formula:- n is the no. of observation / rows

$$\boxed{\text{Degree of Freedom} = n-1}$$

→ This is mostly used when estimating the mean or Variance from sample data.

※ Why does DOF reduces:

When we estimate something from data (like mean), we use one value (mean itself) as a fixed reference.

→ So we lose one free value → that's why we substract 1

The total DOF in LR is divided into 2 components:

i) Degree of Freedom for the model :- (df-model) (k)

This is equal to the no. of independent variables in the model (i)

ii) Degree of freedom for the Residual :- (df-residual)

This tells how many data points are left to estimate error / variance after fitting the model.

→ indicates how many observations still have freedom after fitting the model line / curve ⇒ ~~no~~

→ no. of datapoints(n) − no. of estimated parameters including intercept

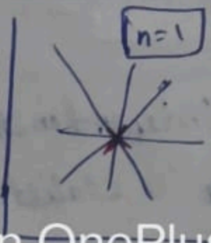
$(n) - (k+1) \Rightarrow \boxed{n-(k+1)}$

$n \Rightarrow$ no. of datapoints
$k+1 \Rightarrow$ estimated parameters (k slopes + 1 intercept)
$k \Rightarrow$ no. of input cols / features

$$\begin{aligned} DOF &= df\text{-}m + df\text{-}r \\ &= k + n-k-1 \\ &= \boxed{n-1} \end{aligned}$$

Simple L.R:- how many data points are required to draw a Regression line — X / Y

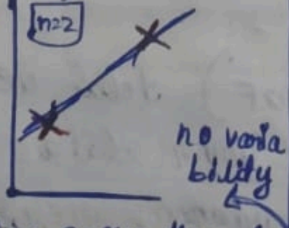
$\boxed{n=1}$



There are too many Regression lines

i) for 1 data point

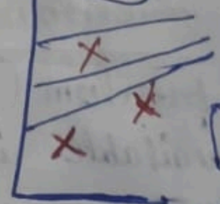
ii) 2 data points $\boxed{n=2}$



no variability

line passes through 2 data points because of

$\boxed{n=3}$



• here it has freedom to vary

$\boxed{n=3}$ $\boxed{k=1}$

$DOF = n-k-1 = 3-1-1 = 1$

Why do we use n-1 in statistics:-

→ when we calculate variance or standard deviation, we first compute mean from the same data.

→ This creates a restriction ⇒ one value is no longer free

Ex:- 5, 8, 10 (computing mean)

⇒ If mean = 7.66, If two values are known, the 3rd value is fixed

⇒ so only n-1 values are free to vary

∴ we divide by n-1 to get an unbaised estimate of population variance

Using Statsmodel [model.summary()]

LR

i) finding whether there is linear correlation -ship b/w X & Y using hypothisis test [F-statistic]

X	Y

ii) if it is a linear relation, findinding the strength b/w X and Y or goodness of it using R2-score & adjusted

iii) $y(x_1, x_2, x_3)$ → analyzing how independent cols effecting the targeted col Y.

F-Statistic & Prob(F-Statistic):-

It is a test conducted to check the relation b/w X & Y exists or not

The F-test for overall significance is a statistical test used to determine whether a linear regression model statistically significant. [meaning it provides a better fit to the data than just using the mean of the dependent variable (Y)

→ F-test regression checks whether your independent variables actually helps to explain the dependent variable.

→ If the F-test is significant, means your overall Reg model is us